

Optical properties

9.1 Absorption from exciting polarons' self-trapped carriers

With polaron formation self-trapped electronic carriers are bound within potential wells established by shifts of atoms' equilibrium positions. Optical absorption attributed to polarons results from photo-excitation of their bound electronic carriers. The frequencies of these absorbed photons typically exceed atomic-vibration frequencies. As a result, one can invoke the Franck–Condon principle and presume that atoms remain fixed during the photo-excitation process (Franck, 1926; Condon, 1926). Thus a polaron's optical absorption simply probes photo-excitation of its self-trapped carrier to its excited electronic energy levels (Emin, 1993). The spectra of excited electronic states for large and small polarons differ qualitatively from one another to produce distinguishable optical absorptions.

A large polaron's self-trapped carrier extends over multiple structural units since its electronic bandwidth exceeds the depth of the potential well which binds it. In these circumstances the self-trapping medium can be treated as a deformable ionic continuum rather than as a solid composed of discrete ions. The eigenvalue spectrum for the self-trapped electron then resembles that for a donor electron in a conventional semiconductor.

As illustrated in Fig. 9.1, the electron can be photo-excited from the ground state of its Coulombic self-trapping potential well to unbound states associated with the conduction band. The threshold for this Franck–Condon absorption is just the electronic portion of the large polaron's binding energy. A large polaron's linear long-range electron–phonon interaction and its self-trapped carrier's Coulombic potential well produce simple numerical relations between contributions to its net binding energy E_p . In particular, net binding results from lowering the ground-state carrier's potential energy by $4E_p$ while increasing the material's strain energy by $2E_p$ and the carrier's confinement energy by half of the net lowering of the potential

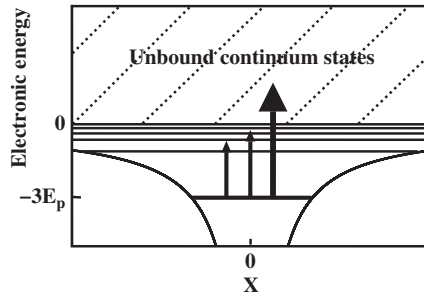


Fig. 9.1 Bound and unbound electronic energy levels of a large-polaron's self-trapped carrier are depicted as horizontal lines and a hatched area, respectively. Absorption processes for a large polaron include photo-excitation of its self-trapped carrier from its ground state (thick horizontal line) (1) to unbound continuum states (thick vertical arrow) and (2) to excited bound states (thinner vertical arrows).

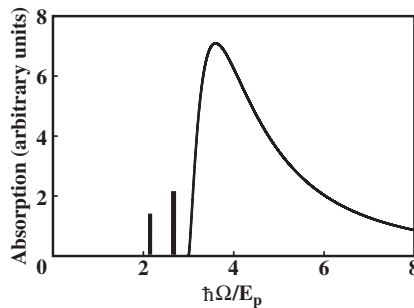


Fig. 9.2 A large polaron's absorption coefficient (in arbitrary units) is plotted against the energy of the exciting photon in units of the large-polaron binding energy, E_p . Photo-excitation of the self-trapped carrier to unbound continuum states produces a broad asymmetric absorption band. Photo-excitations of the self-trapped carrier to excited bound states produce relatively narrow absorption bands below the threshold energy $3E_p$.

energy $(4E_p - 2E_p)/2 = E_p$ (Emin, 1993). Therefore the threshold for photo-dissociation of the large polaron, the electronic portion of its binding energy, is $3E_p$.

A photon energy $\hbar\Omega$ above this threshold imparts kinetic energy of $\hbar\Omega - 3E_p$ to the freed electron (Emin, 1993). Moreover, the probability of exciting a self-trapped electron into the conduction band (1) increases with its density-of-states and (2) decreases rapidly when the wavevector of the freed electron exceeds $1/R_{LP}$, the reciprocal of the radius of the self-trapped electron. Thus, as illustrated in Fig. 9.2, the absorption coefficient arising from freeing a large polaron's self-trapped carrier is an asymmetric peaked function of photon energy whose width is typically much less than that of the conduction band.

Photons with energies below the threshold energy for liberating self-trapped carriers can also be absorbed. As illustrated in Fig. 9.1, photo-absorption can elevate a self-trapped carrier to an excited bound state of its self-trapping potential well. As depicted in Fig. 9.1, some of the potential photo-excitation processes may be forbidden by symmetry-based selection rules.

A small polaron's self-trapped carrier collapses to a single site with its binding energy exceeding its electronic bandwidth. In these circumstances, it is expedient to treat a small polaron as localized at one of a solid's discrete sites (Emin, 1993). As illustrated in Fig. 9.3, absorption of a photon can excite this self-trapped carrier from a low-lying state on one site to a higher energy state on a different site. In addition, a photon of equal energy can be emitted in the reverse process: transferring a carrier from the high-energy state on one site to a lower-lying state on a different site. The photon-absorption process dominates in thermal equilibrium since the carrier is then more likely to be found in the low-energy configuration than in the high-energy configuration.

For vanishing photon energy, $\Omega = 0$, the net absorption is simply proportional to the small-polaron's dc electrical conductivity (Klinger, 1963; Reik and Heese, 1967; Emin, 1975a). As the photon energy is increased the net absorption rises for the strong electron–phonon coupling characteristic of small-polaron formation. The absorption reaches its peak when the photon's energy equals the electronic energy needed to promote a small-polaron's self-trapped carrier from its ground state to an adjacent site while (in accord with the Franck–Condon principle) freezing atoms' positions.

The absorption peak's width is produced by atoms' departures from their ground-state equilibrium positions. As such, the absorption peak is most narrow at very low temperatures where atoms' zero-point motions predominate. As illustrated in Fig. 9.4 a small-polaron absorption band broadens with increasing temperature.

With small-bipolaron formation two carriers share a common site. Each carrier experiences an equilibrated self-trapping potential well that is twice as deep as that for an equilibrated small polaron's single carrier. Thus the net electronic energy of the Holstein small-bipolaron of Section 7.2 becomes $-8E_b + U$, where U is the two

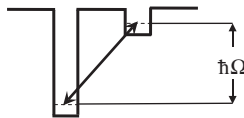


Fig. 9.3 A small-polaron's photo-absorption usually transfers its severely localized carrier from a state at one site to a higher lying state at a neighboring site. Photons also can be emitted whenever a localized carrier can be transferred from a high-energy state on one site to a low-lying state on an adjacent site.

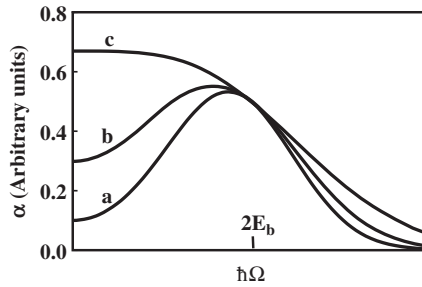


Fig. 9.4 A typical small-polaron absorption coefficient is plotted versus photon energy $\hbar\Omega$ for three temperatures. (a) At low temperature the absorption forms a band that peaks when $\hbar\Omega$ nears the difference between the electronic energy at an equilibrated unoccupied neighboring site and the electronic energy of the equilibrated self-trapped carrier. In the Holstein model (Holstein, 1959b) this energy is $2E_b$, defined below Eq. (7.6). (b) At higher temperatures increased displacements of atoms from their equilibrium positions broaden the absorption band. (c) At very high temperatures broadening obscures the peaked behavior.

carriers' mutual on-site Coulomb repulsion. After optically transferring one of the small-bipolaron's carriers to an equilibrated unoccupied site the system's electronic energy becomes $-4E_b$. The difference between the final and initial electronic energies is the energy of the exciting photon $\hbar\Omega = -4E_b - (-8E_b + U) = 4E_b - U$. Since formation of a conventional small bipolaron requires that $2E_b > U$, the energy of its characteristic photon, the energy of its absorption peak, is always greater than that for a small polaron, $2E_b$: $4E_b - U > 2E_b$ for $2E_b > U$. By contrast, no absorption is possible if $U > 4E_b$. This circumstance occurs if the bipolaron is a "softening" bipolaron, stabilized by carrier-induced reduction of vibration frequencies rather than by carrier-induced displacements of atoms' equilibrium positions (Emin, 2000b).

9.2 Low-frequency absorption from polaron motion

In addition to absorption that excites polarons' self-trapped carriers, polarons' slow motion enables low-frequency absorption. Large-polarons' coherent motion produces a Drude-like absorption that falls with increasing frequency. Small polarons' phonon-assisted hopping in disordered media produces an absorption that typically rises with increasing frequency.

A large polaron moves very slowly with a very large effective mass, Eq. (4.39). In particular, a large polaron's movement follows that of the atoms responsible for its carrier's self-trapping. The concomitant large-polaron bandwidth, Eq. (4.40), is less than the phonon energy characterizing vibrations of the atoms responsible for self-trapping. As a result, conservation of energy for a large polaron precludes the

one-phonon emission and absorption processes which dominate the scattering of a conventional carrier. Rather a large polaron and phonons scatter via a two-phonon process (Schüttler and Holstein, 1986). This scattering occurs as long-wavelength ($> R_{LP}$) phonons experience the region of softened vibrations surrounding a large polaron. With sufficiently dispersive phonons a large polaron “reflects” the long-wavelength ($> R_{LP}$) phonons it encounters. The transport relaxation time τ for this process is given by

$$\frac{1}{\tau} \approx \frac{\kappa T}{m_{LP} \omega R_{LP}^2} \approx \omega \left(\frac{\kappa T}{E_p} \right), \quad (9.1)$$

when the thermal energy exceeds the energy of these phonons, $\kappa T > \hbar \omega$. As a result, the Drude electrical conductivity resulting from coherent motion of these large polarons,

$$\sigma(\Omega) = \frac{\sigma(0)}{1 + (\Omega\tau)^2}, \quad (9.2)$$

is distinguished from that of conventional carriers by its extremely long relaxation time τ . The very long large-polaron relaxation time indicates the difficulty of phonons’ scattering these very massive quasi-particles.

Disorder often accompanies small-polaron and small-bipolaron formation. Disorder is intrinsic in semiconducting glasses and is produced extrinsically by imbedding dopants, impurities or defects within crystalline semiconductors. At low enough temperatures dc transport is limited by difficult thermally assisted hops that transport carriers between localized regions of relatively easy movement. By contrast, carriers that move with minimal thermal assistance within these regions of relatively easy transport dominate the low-temperature frequency-dependent electrical conductivity. The net polarization conductivity is the sum of contributions from these regions. The real part of the complex conductivity at frequency Ω produced by a carrier restricted to a region denoted by index i is:

$$\text{Re}[\sigma_i(\Omega)] \propto \Omega \left[\frac{2\Omega\tau_i}{1 + (\Omega\tau_i)^2} \right], \quad (9.3)$$

where τ_i is the relaxation time for the carrier’s polarization current. The square-bracketed term on the right-hand side of Eq. (9.3) peaks when $\Omega\tau_i = 1$. Thus the observation of a low-frequency ($< 10^6$ Hz) polarization conductivity indicates the presence of relatively conducting regions with slow relaxation, large values of τ_i .

The classic case of low-frequency hopping conductivity is that of low-temperature electron hopping between shallow impurities in doped silicon. Specifically, low-frequency (10^2 – 10^5 Hz) conductivity is observed for the low-temperature (1–5 K) hopping of charge carriers between very low densities (10^{15} – 10^{17} cm $^{-3}$) of shallow impurities in compensated crystalline silicon (Pollak and Geballe, 1961). The very slow relaxation of the ac conductivity is attributed to long ($\gg 100$ Å) hops between isolated pairs of shallow impurities. In particular, the small electronic overlap between the sites involved in very long hops gives them very small inter-site electronic transfer energies. With sufficiently small electronic transfer energies ($< \hbar\omega$) hops become “non-adiabatic”, cf. Chapter 11. The concomitant hopping rates are very small since they are proportional to the square of their electronic transfer energies. These long hops dominate the low-frequency ac conductivity of lightly doped compensated silicon.

Low-frequency conductivity is also observed for low-temperature small-polaron and small-bipolaron hopping (Austin and Mott, 1969; Samara *et al.*, 1993). In these instances the hopping distances are comparable to a lattice constant (< 5 Å). Indeed, the electronic transfer energies are often large enough ($> \hbar\omega$) to support adiabatic rather than non-adiabatic hopping. The magnitudes of the transfer energies are then not paramount in determining carriers’ hopping rates. Rather, small-polarons’ or small-bipolarons’ self-trapped carriers move very slowly at low temperatures because their motion requires significant movement of the atoms associated with their self-trapping (Emin, 1992). By contrast, a hop of a carrier between shallow impurity-states requires only minimal atomic motion since atomic relaxation surrounding large-radius states is relatively small.

Disorder causes the relaxation times from different polarization centers to diverge from one another. Dispersion of the relaxation times for low-temperature hopping between shallow impurities results primarily from a spread of hopping distances. Dispersion of the rates for adiabatic small-polaron and small-bipolaron hopping results from a spread of energy differences between the involved sites. Low-temperature small-polaronic hops between nearly degenerate sites are especially sluggish due to the very low density of low-energy acoustic phonon states (Emin, 1992). With sufficient dispersion of the rates governing slow hops, the net conductivity typically increases sublinearly with increasing frequency at very low frequencies: $\text{Re}[\sigma(\Omega)] \propto \Omega^{1-s}$ (Pollak and Pike, 1972). This behavior is widely reported for hopping transport in disordered materials. However, these observations by themselves usually establish neither the hopping mechanism nor the source of relaxation rates’ dispersion.

9.3 Carriers' photo-generation, recombination and luminescence

In conventional high-mobility semiconductors electrons and holes produced by the absorption of super-bandgap photons generally separate and thereby contribute to the dc conductivity. Photo-conduction ends as carriers become trapped. Ultimately carriers of opposing signs reunite via recombination centers, defects or impurities at which the oppositely signed carriers are sequentially captured.

A very different situation often prevails when both photo-generated carriers move slowly enough that they collapse into small polarons. Then only a small fraction of the electron-hole pairs generated from weakly absorbed photons separate from one another to contribute to photo-conduction. Rather the unseparated carriers ultimately merge with their nascent partners in a process termed geminate recombination. These well-localized carriers often recombine without the assistance of extrinsic traps, the recombination centers required in conventional high-mobility semiconductors.

The probability of separating a photo-induced slow moving electron and hole in a semiconductor is often treated as analogous to the thermal dissociation of oppositely charged ions in a weak electrolyte (Onsager, 1934; 1938). The probability of separating the electron and hole is then expressed as $\exp(-R_c/s_0)$. Here $R_c \equiv e^2/\epsilon_0\kappa T$ defines the carriers' Coulomb capture radius and s_0 denotes the characteristic separation of thermalized photo-generated carriers, a strongly increasing function of photo-excitation energy. With $R_c \gg s_0$ most of these photo-generated electron-hole pairs geminately recombine. Applying an electric field assists a pair's dissociation by driving its electron and hole in opposing directions. The carrier generation efficiency in an applied electric field of strength F is then written as $\exp(-R_c/s_0)[1 + qR_cF/2\kappa T]$. The parameters R_c and s_0 can be obtained from measurements of the slope and the intercept at $F = 0$ of the carrier-generation efficiency plotted versus the applied electric field. Small radiation-induced carrier-generation efficiencies and geminate recombination are reported in diverse unconventional semiconducting materials. Some examples are: anthracene, PVK (poly-N-vinylcarbazole) and amorphous selenium (Kepler and Coppage, 1966; Batt, Braun and Hornig, 1968; Hughes, 1971a; 1971b; Melz, 1972; Pai and Enck, 1975; Chance and Braun, 1976).

With sufficiently slow motion these separated pairs of carriers will self-trap to form oppositely charged small polarons. Oppositely charged small polarons interact directly through their Coulomb attraction and indirectly through interference of their atomic-displacement patterns. Interference occurs as both carriers strive to displace each of the material's atoms. Such interference causes the net energy of the two oppositely charged small polarons to depend on their separation. Destructive interference causes the net energy of the two oppositely charged small polarons to rise as their separation is reduced. The sum of this repulsion and the long-range

Coulomb attraction of two oppositely charged polarons is plotted against their separation in Fig. 9.5. This illustration shows the total energy having its minimum at a finite separation.

Photo-generated species also include excitations in which electron and hole do not separate from one another. A mutually bound pair of oppositely signed carriers that share a common spatial region is regarded as an *exciton*. A Wannier exciton has at least one of its two oppositely charged carriers extending among multiple sites. A Frenkel exciton has both carriers sharing a common structural unit (e.g. a molecule or a bond). The binding of both types of exciton is enhanced by the relaxation of surrounding atoms. Since an exciton has no net charge, it has minimal Coulomb interactions with distant ions. Thus, atomic relaxation about an exciton is primarily governed by the electron–phonon interaction’s short-range component rather than its long-range (Coulomb) component. An exciton together with the atomic relaxation that accompanies it is termed a *self-trapped exciton*. Figure 9.5 indicates the energy of an exciton and that of the self-trapped exciton which forms after atoms relax about it.

Luminescence at progressively lower energies results as the electrons and holes of excitons, self-trapped excitons and oppositely charged small polarons recombine with the emission of photons. The recombination of an exciton’s electron and hole tends to be relatively rapid because their overlap is relatively strong. Lower-energy luminescence from radiative recombination of self-trapped excitons tends to be postponed since their formation is delayed by the necessity of transcending their energy barriers to self-trapping described in Section 4.3. Still lower-energy

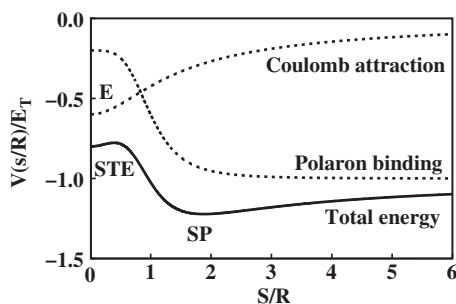


Fig. 9.5 The energy of two oppositely charged small polarons $V(s/R)$ is plotted (solid line) in units of their net binding energy E_T against their separation s divided by a small-polaron radius R . This energy sums the two polarons’ Coulomb attraction with their polaron binding (dashed curves). Repulsion occurs as oppositely charged small polarons compete to displace surrounding atoms. The energy minima correspond to a self-trapped exciton (STE) and to separated small-polarons (SP). The minimum of the Coulomb attraction at $s/R = 0$ corresponds to the formation of an exciton (E).

luminescence from direct recombination of an electron-and-hole small-polaron pair tends to be slowest since their overlap with one another is relatively weak.

The luminescence spectrum from direct recombination of electron-and-hole small polarons evolves as their phonon-assisted hopping alters their separations. In particular, small-polarons' hopping facilitates their converging on their minimum-energy separations. In addition, as discussed in Section 9.1 and depicted in Fig. 9.3, irradiation within small-polarons' absorption bands induces small polarons' hopping. Indeed, small-polaron optical absorption bands are bleached out when absorption within these bands facilitates small-polarons' elimination via their recombination

Recombination occurs via phonon emission as well as via photon emission. These non-radiative and radiative recombination processes are illustrated using the generic configuration diagram of Fig. 9.6 (Dexter *et al.*, 1955). The upper curve depicts the energy of a deformable medium containing an electronic excitation plotted against x , a deformation parameter upon which it depends. In particular, the electronic excitation can be an exciton or separated electron-and-hole small polarons. The lower curve depicts the ground-state energy's harmonic dependence on this deformation parameter. The upward arrow at $x = 0$ represents photo-absorption of energy Δ_e by the undeformed ground state creating an electronic excitation. The subsequent displacements of atoms surrounding the electronic excitation correspond to the energy of the upper curve relaxing to its minimum at $x = x_m$. Luminescence of energy Δ_l from the excited-state minimum is represented by the downward arrow at $x = x_m$. The transition from the upper curve to the lower curve via their crossing point at energy E_{nr} and $x = x_{nr}$ corresponds to non-radiative recombination for classical atomic motion. With (1) a short-range linear electron–phonon interaction characterized by the force F and (2) both curves varying harmonically with stiffness constant k : $x_m = F/k$, $E_m = \Delta_e - F^2/2k$, $\Delta_l = \Delta_e - F^2/k$, $x_{nr} = \Delta_e/F$ and $E_{nr} = \Delta_e^2/(2F^2/k)$.

Recombination is a competition between radiative and non-radiative processes. Non-radiative recombination generally increases with electron–phonon interactions' strength. Luminescence results when electron–phonon interactions are ineffective in producing non-radiative recombination. This framework is adopted to explain why luminescence is observed from some F -centers but is not detected from others (Bartram and Stoneham, 1975).

The difference between the characteristic absorption energy and the luminescence energy is termed the luminescence Stokes shift. This energy generally heats a solid. In some instances (e.g. in alkali halides) this energy is imparted to atoms in such a way as to also efficiently produce structural defects (Pooley, 1966a; 1966b).

Irradiation that cools a material can also be envisioned (Pringsheim, 1929; Landau, 1946). As illustrated in Fig. 9.6, emission of a photon of energy Δ_l after

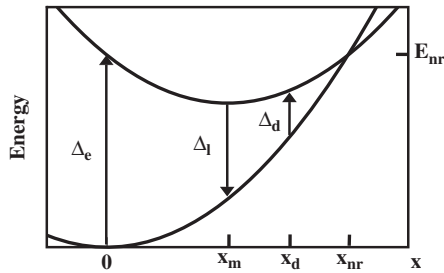


Fig. 9.6 Sums of the strain energy and the electronic energy for the electronic ground state (lower curve) and for an excited state (upper curve) are plotted against the deformation parameter x . The photo-excitation energy from the undeformed ground state is Δ_e . The energy of luminescence from the excited-state minimum at $x = x_m$ is Δ_l . The photo-excitation energy when the ground state is deformed to $x = x_d$ is Δ_d . The crossing point of the two curves at $x = x_{nr}$ establishes E_{nr} , the characteristic energy for a semiclassical non-radiative transition between the two states.

creation of an excitation by photo-exciting a deformed ground state with a photon of energy Δ_d extracts energy $\Delta_l - \Delta_d$ from the material. However, realizable optical cooling schemes require electron–phonon interactions that are sufficiently ineffective so that heating from ancillary Stokes shifts and non-radiative recombination are exceptionally small (Epstein *et al.*, 1995; Emin, 2007).

Optical properties of amorphous chalcogenide films appear consistent with the expectations of materials whose carriers form small polarons (Emin, 1980a). In particular, the observation that super-bandgap irradiation produces significant super-bandgap luminescence implies that relaxation of these electronic excitations is impeded by rapid localization (Shah and Bösch, 1979). Indeed, optically induced carriers manifest germinate recombination (Pai and Enck, 1975). Furthermore, time-resolved measurements find that optically induced carriers move with extremely low mobilities and recombine very slowly (Fork *et al.*, 1979). In addition, photo-conductivity measurements indicate an electron mobility that is too small to measure (Ing *et al.*, 1971). These results are consistent with steady-state charge transport being dominated by hole small polarons (Emin, Seager, and Quinn, 1972; Seager and Quinn, 1975). The nature of these carriers is suggested by the observation that super-bandgap excitations produce electron paramagnetic resonance (EPR) signals attributed to electrons on arsenic atoms and holes on chalcogen atoms (Bishop *et al.*, 1977). Super-bandgap irradiation also generates a mid-gap absorption band which together with the induced EPR signals can be bleached out with mid-gap irradiation (Bishop *et al.*, 1977). Recombination is primarily non-radiative with steady-state super-bandgap illumination producing a large luminescence Stokes shift of about half the semiconducting energy gap (Fischer *et al.*, 1971). Time-dependent

luminescence measured after pulsed illumination reveals several bands that evolve to leave just the steady-state result at long times (Bösch and Shah, 1979). The time dependences of the three observed luminescence bands has been described in terms of exciton, self-trapped exciton and small-polaron radiative recombination (Bösch *et al.*, 1980). An alternative model for the electrical transport and optical properties of chalcogenide glasses proposes that they result from polaronic electron and hole traps associated with high densities of bonding defects (Street and Mott, 1975; Kastner *et al.*, 1976).

These findings are similar to those from transition-metal compounds in which charge carriers are thought to form small polarons. In particular, steady-state transport measurements on LiNbO_3 indicate that electrons form small polarons (Nagels, 1980). Moreover, super-bandgap irradiation produces a mid-gap absorption band and two EPR signals from two distinct types of localized spins, O^- and Nb^{3+} ions. (Schirmer and von der Linde, 1978; Schirmer *et al.*, 2009).